

PE-EC801C — Error Correcting Codes

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Module 2: Galois Fields and Cyclic Codes

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1. Algebraic Background — Groups, Rings and Fields

To design and analyze cyclic, BCH, and Reed-Solomon codes, we must establish their underlying algebraic structures.

Algebraic Structures Summary

- **Group:** A set G with a binary operation $*$ satisfying: closure ($a * b \in G$), associativity ($a * (b * c) = (a * b) * c$), existence of identity e ($a * e = a$), and existence of inverse a^{-1} ($a * a^{-1} = e$). If $a * b = b * a$, it is an **Abelian Group**.
- **Ring:** A set R with two binary operations (addition $+$ and multiplication $\cdot$) where $(R, +)$ is an Abelian group, multiplication is associative and distributive over addition, and contains a multiplicative identity 1.
- **Field:** A ring F where multiplication is commutative, and every non-zero element $a \in F$ has a multiplicative inverse a^{-1} such that $a \cdot a^{-1} = 1$.

1.1 Finite Fields (Galois Fields)

A field containing a finite number of elements is a **Galois Field**, denoted as $\text{GF}(q)$. A finite field exists if and only if q is a prime power: $q = p^m$, where p is a prime number (called the **characteristic** of the field) and m is a positive integer.

- For $m = 1$, $\text{GF}(p)$ is the prime field consisting of integers $\{0, 1, \dots, p-1\}$ under modulo- p arithmetic.
- For $m > 1$, $\text{GF}(2^m)$ is an extension field constructed using polynomials.

1.2 Extension Field Construction

To construct $\text{GF}(2^m)$, we choose an **irreducible polynomial** $p(x)$ (a polynomial that cannot be factored into lower-degree polynomials over $\text{GF}(2)$) of degree m . Let α be a root of $p(x)$, such that $p(\alpha) = 0$. The elements of $\text{GF}(2^m)$ can be represented in three ways: 1. **Power representation:** $\{0, \alpha^0, \alpha^1, \dots, \alpha^{2^m-2}\}$. 2. **Polynomial representation:** Binary combinations of $\{1, \alpha, \alpha^2, \dots, \alpha^{m-1}\}$. 3. **Vector representation:** Binary coefficients of the polynomial representation.

Construction of $\text{GF}(2^3)$ using $p(x) = x^3 + x + 1$

Let α be a root of $p(x)$ $\implies \alpha^3 \oplus \alpha \oplus 1 = 0 \implies \alpha^3 = \alpha \oplus 1$. The $2^3 = 8$ elements of $\text{GF}(8)$ are:

Power Form	Polynomial Form	Vector Form (a_2, a_1, a_0)
0	0	(0, 0, 0)
α^0	1	(0, 0, 1)
α^1	α	(0, 1, 0)
α^2	α^2	(1, 0, 0)
α^3	$\alpha \oplus 1$	(0, 1, 1)
α^4	$\alpha^2 \oplus \alpha$	(1, 1, 0)
α^5	$\alpha^2 \oplus \alpha \oplus 1$	(1, 1, 1)
α^6	$\alpha^2 \oplus 1$	(1, 0, 1)

1.3 Primitive Elements and Minimal Polynomials

- **Primitive Element:** An element $\beta \in \text{GF}(2^m)$ whose successive powers generate all $2^m - 1$ non-zero elements of the field.
- **Conjugate Elements:** If $\beta \in \text{GF}(2^m)$, its conjugates over $\text{GF}(2)$ are:

$$\beta, \beta^2, \beta^{2^2}, \dots, \beta^{2^{d-1}}$$

where d is the smallest integer such that $\beta^{2^d} = \beta$.

- **Minimal Polynomial $(m(x))$:** The lowest-degree polynomial with binary coefficients that has β (and its conjugates) as a root:

$$m(x) = \prod_{i=0}^{d-1} (x - \beta^{2^i})$$

2. Factorization of $X^n - 1$ over a Finite Field

Let n be the block length of a cyclic code. The polynomial $X^n \oplus 1$ can be factored into a product of irreducible polynomials over $\text{GF}(2)$:

$$X^n \oplus 1 = f_1(x)f_2(x) \dots f_r(x)$$

The roots of these irreducible factors are elements of $\text{GF}(2^m)$, where m is the multiplicative order of 2 modulo n . Every irreducible factor $f_i(x)$ is the minimal polynomial of a root of $X^n \oplus 1$.

Factorization Example for $n = 7$

We wish to factor $X^7 \oplus 1$ over $\text{GF}(2)$. The roots of $X^7 \oplus 1 = 0$ are the 7-th roots of unity in $\text{GF}(8)$, which are $\{1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6\}$. Grouping roots into conjugate classes (cyclotomic cosets):

- Coset 1: $\{1\}$ (minimal polynomial: $x \oplus 1$)
- Coset 2: $\{\alpha, \alpha^2, \alpha^4\}$ (minimal polynomial: $x^3 \oplus x \oplus 1$)
- Coset 3: $\{\alpha^3, \alpha^6, \alpha^5\}$ (minimal polynomial: $x^3 \oplus x^2 \oplus 1$)

Therefore, the factorization of $X^7 \oplus 1$ over $\text{GF}(2)$ is:

$$X^7 \oplus 1 = (x \oplus 1)(x^3 \oplus x \oplus 1)(x^3 \oplus x^2 \oplus 1)$$

3. Cyclic Codes

A **cyclic code** is a linear block code where any cyclic shift of a codeword is also a valid codeword.

Formal Cyclic Shift Property

If $c = (c_0, c_1, \dots, c_{n-1}) \in C$ is a codeword, then its cyclic shift:

$$c^{(1)} = (c_{n-1}, c_0, c_1, \dots, c_{n-2}) \in C$$

is also a valid codeword.

3.1 Polynomial Representation

Codewords are represented as polynomials of degree at most $n - 1$:

$$c(x) = c_0 \oplus c_1x \oplus c_2x^2 \oplus \dots \oplus c_{n-1}x^{n-1}$$

A cyclic shift corresponds to multiplying $c(x)$ by x modulo $X^n \oplus 1$:

$$x \cdot c(x) \pmod{X^n \oplus 1} = c_{n-1} \oplus c_0x \oplus c_1x^2 \oplus \dots \oplus c_{n-2}x^{n-1}$$

Algebraically, a cyclic code of length n is an **ideal** in the quotient polynomial ring $\text{GF}(2)[x]/(X^n \oplus 1)$.

3.2 The Generator Polynomial $g(x)$

A cyclic code is uniquely defined by a single generator polynomial $g(x)$.

Properties of the Generator Polynomial

- $g(x)$ is the unique, monic polynomial of minimum degree in the code.
- The degree of $g(x)$ is equal to the number of parity check bits: $\deg(g(x)) = n - k$.
- $g(x)$ is a factor of $X^n \oplus 1$:

$$X^n \oplus 1 = g(x) \cdot h(x)$$

where $h(x)$ is the **parity check polynomial** of degree k .

Every codeword polynomial $c(x)$ in the code can be written as a product of a message polynomial $m(x)$ (of degree $< k$) and the generator polynomial $g(x)$:

$$c(x) = m(x) \cdot g(x)$$

3.3 Systematic Encoding Algorithm

To construct a systematic cyclic code where the message bits occupy the higher-order coefficients of $c(x)$, we use the following steps:

Systematic Cyclic Encoding Steps

Step 1: Multiply the message polynomial $m(x)$ by x^{n-k} to shift it into the systematic position.

Step 2: Divide $x^{n-k}m(x)$ by the generator polynomial $g(x)$ to find the remainder $d(x)$:

$$x^{n-k}m(x) = q(x)g(x) \oplus d(x)$$

where $\deg(d(x)) < n - k$.

Step 3: Construct the codeword polynomial:

$$c(x) = x^{n-k}m(x) \oplus d(x) = q(x)g(x)$$

Since addition and subtraction are identical in binary (modulo-2), adding $d(x)$ to $x^{n-k}m(x)$ cancels the remainder, leaving a polynomial that is a multiple of $g(x)$ (hence a valid codeword).

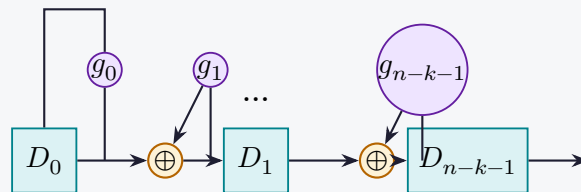
4. Feedback Shift Register Encoders and Syndrome Calculators

The polynomial division needed for systematic cyclic encoding is implemented in hardware using a **Feedback Shift Register (FSR)**.

4.1 Systematic FSR Encoder Circuit

Let $g(x) = g_0 \oplus g_1x \oplus \dots \oplus g_{n-k-1}x^{n-k-1} \oplus x^{n-k}$ be the generator polynomial. The shift register consists of $n - k$ stages.

Systematic Cyclic Encoder Circuit



The division operates as follows:

1. The gate switch is closed. The k message bits are fed into the channel and the feedback path simultaneously.
2. After k clock cycles, the shift registers contain the remainder coefficients (parity bits).
3. The gate switch is opened, disabling feedback, and the parity bits are shifted out to complete the n -bit systematic codeword.

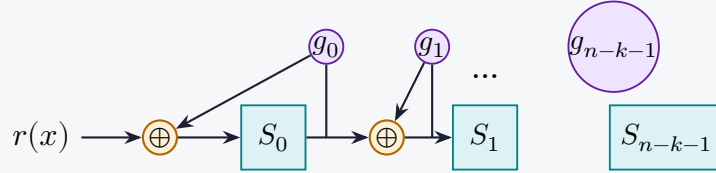
4.2 FSR Syndrome Calculator Circuit

The syndrome $S(x)$ of a received polynomial $r(x)$ is the remainder of $r(x)$ divided by $g(x)$:

$$S(x) = r(x) \pmod{g(x)}$$

This is implemented using a feedback shift register similar to the encoder.

Syndrome Calculator Circuit



5. Spectral Properties of Cyclic Codes

Like time-domain signals, codewords of cyclic codes have equivalent frequency-domain descriptions using the **Discrete Fourier Transform (DFT)** over finite fields. Let $v = (v_0, v_1, \dots, v_{n-1})$ be a vector over $\text{GF}(2^m)$ where n divides $2^m - 1$. Let α be a primitive n -th root of unity in $\text{GF}(2^m)$.

Finite Field Discrete Fourier Transform

The DFT of v , denoted $V = (V_0, V_1, \dots, V_{n-1})$, consists of elements in $\text{GF}(2^m)$ computed by:

$$V_j = \sum_{i=0}^{n-1} v_i \alpha^{ij}, \quad j = 0, 1, \dots, n-1$$

The inverse DFT is:

$$v_i = \sum_{j=0}^{n-1} V_j \alpha^{-ij}, \quad i = 0, 1, \dots, n-1$$

5.1 Spectral Constraints and Ideals

The cyclic shift of a vector v in the time domain corresponds to multiplying its spectral components by powers of α :

$$v^{(1)} \longleftrightarrow (\alpha^0 V_0, \alpha^1 V_1, \dots, \alpha^{n-1} V_{n-1})$$

Therefore, a cyclic code can be defined in the spectral domain by forcing certain spectral components to zero:

$$C = \{c \in \text{GF}(2)^n \mid C_j = 0 \text{ for all } j \in \mathcal{K}\}$$

where $\mathcal{K}$ is a set of spectral indices. This spectral property is the foundation for designing BCH and Reed-Solomon codes.

Quick Revision — Module 2

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Key Formulas and Metrics at a Glance

Galois Field order	$q = p^m$ (p is prime characteristic)
Roots of Unity factor	$X^n \oplus 1 = g(x) \cdot h(x)$
Generator degree	$\deg(g(x)) = n - k$
Systematic Codeword polynomial	$c(x) = x^{n-k}m(x) \oplus [x^{n-k}m(x) \pmod{g(x)}]$
Modulo-2 addition relation	$a(x) \oplus b(x) = a(x) - b(x)$
Parity relation	$g(x) \cdot h(x) \equiv 0 \pmod{X^n \oplus 1}$
Finite Field DFT spectrum	$V_j = \sum_{i=0}^{n-1} v_i \alpha^{ij}$
Conjugate elements of β	$\beta^{2^0}, \beta^{2^1}, \beta^{2^2}, \dots, \beta^{2^{d-1}}$
Generator matrix polynomial	$G = \begin{bmatrix} g(x) \\ xg(x) \\ \vdots \\ x^{k-1}g(x) \end{bmatrix}$ in non-systematic form

One-Line Definitions

- **Abelian Group:** A group where the binary operation is commutative ($a * b = b * a$).
- **Field:** An algebraic ring where multiplication is commutative and every non-zero element has a multiplicative inverse.
- **Galois Field:** A finite field containing a prime-power number of elements, denoted $\text{GF}(p^m)$.
- **Irreducible Polynomial:** A polynomial that cannot be factored into polynomials of lower degree over the same field.
- **Primitive Element:** An element whose powers generate all non-zero elements of a finite field.
- **Minimal Polynomial:** The lowest-degree binary polynomial having a field element (and its conjugates) as roots.
- **Cyclic Code:** A linear code where a cyclic shift of any codeword produces another valid codeword.
- **Generator Polynomial ($g(x)$):** The unique lowest-degree polynomial of a cyclic code, which divides $X^n \oplus 1$.

Frequently Asked / Viva Questions

Q1. What are the differences between groups, rings, and fields?

A group has one operation (+). A ring has two operations (+ and $\cdot$) and is an Abelian group under addition. A field is a commutative ring with a multiplicative identity where every non-zero element has a unique multiplicative inverse.

Q2. Why do Galois Fields only exist for prime-power sizes p^m ?

For fields of size q , the characteristic must be a prime p . The field forms a vector space over its prime subfield $\text{GF}(p)$ of dimension m . Therefore, the number of elements in the field is $p \times p \times \cdots \times p = p^m$.

Q3. How is systematic cyclic encoding performed algebraically?

First, the message $m(x)$ is multiplied by x^{n-k} to shift it. Then, we find the remainder $d(x)$ of $x^{n-k}m(x)$ divided by $g(x)$. The systematic codeword polynomial is constructed as $c(x) = x^{n-k}m(x) \oplus d(x)$.

Q4. Why must the generator polynomial $g(x)$ divide $X^n \oplus 1$?

A cyclic code of length n is an ideal in the polynomial ring modulo $X^n \oplus 1$. The ideal is generated by $g(x)$. Since the identity codeword $0 \equiv X^n \oplus 1 \pmod{X^n \oplus 1}$ must be a multiple of $g(x)$, $g(x)$ must divide $X^n \oplus 1$.

Q5. Explain how a feedback shift register circuit performs polynomial division.

FSRs use shift register stages as memory and XOR gates as modulo-2 adders. Feedback paths multiply register states by coefficients g_i . Feeding coefficients through the loop matches polynomial long division.

Q6. What are conjugate elements in a finite field?

For an element $\beta \in \text{GF}(2^m)$, its conjugates over $\text{GF}(2)$ are its repeated squarings: $\{\beta, \beta^2, \beta^4, \dots, \beta^{2^{d-1}}\}$, where d is the cycle length. They share the same minimal polynomial.

Q7. What is the parity check polynomial $h(x)$ of a cyclic code?

The parity check polynomial is defined by $h(x) = (X^n \oplus 1)/g(x)$. It has degree k and satisfies $c(x) \cdot h(x) \equiv 0 \pmod{X^n \oplus 1}$ for any codeword $c(x)$.

Q8. What are the advantages of cyclic codes over general linear block codes?

Cyclic codes have rich algebraic properties allowing systematic design (like BCH codes). In hardware, encoding and syndrome calculations are implemented using simple feedback shift registers.

Q9. What is a primitive polynomial?

A primitive polynomial of degree m is an irreducible polynomial over $\text{GF}(2)$ that has a primitive element of $\text{GF}(2^m)$ as a root. Its roots generate the entire multiplicative group.

Q10. Explain the spectral definition of cyclic codes.

Using the DFT over finite fields, cyclic shifts correspond to multiplying spectral elements by powers of α . A cyclic code is defined by forcing specific spectral elements of the codewords to zero.

Comparison of Linear Block Codes and Cyclic Codes

Property	Linear Block Codes	Cyclic Codes
Mathematical Model	Vector Spaces over $\text{GF}(2)$	Ideals in the ring $\text{GF}(2)[x]/(X^n \oplus 1)$
Defining Matrices	Generator G and Parity H	Generator $g(x)$ and Parity $h(x)$ polynomials
Encoding Hardware	Matrix multiplication circuits	Feedback Shift Registers (FSR)
Error Correction	Systematic bounding	Rich algebraic code design (BCH/RS)